



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 1

1. Three field quantities are given by:

$$\mathbf{P} = 2 \mathbf{a}_x - \mathbf{a}_z$$

$$\mathbf{Q} = 2 \mathbf{a}_x - \mathbf{a}_y + 2 \mathbf{a}_z$$

$$\mathbf{R} = 2 \mathbf{a}_x - 3 \mathbf{a}_y + \mathbf{a}_z$$

Determine the following:

- $(\mathbf{P} + \mathbf{Q}) \times (\mathbf{P} - \mathbf{Q})$.
- $\mathbf{Q} \cdot \mathbf{R} \times \mathbf{P}$
- $\mathbf{P} \cdot \mathbf{Q} \times \mathbf{R}$
- $\sin \theta_{QR}$
- $\mathbf{P} \times (\mathbf{Q} \times \mathbf{R})$
- A unit vector perpendicular to both $\mathbf{Q}$ and $\mathbf{R}$
- The component of $\mathbf{P}$ along $\mathbf{Q}$

2. $\mathbf{E}$ and $\mathbf{F}$ are vector fields given by $\mathbf{E} = 2x \mathbf{a}_x + \mathbf{a}_y + yz \mathbf{a}_z$ and $\mathbf{F} = xy \mathbf{a}_x - y^2 \mathbf{a}_y + xyz \mathbf{a}_z$.

Determine:

- $|\mathbf{E}|$ at (1, 2, 3).
- The component of $\mathbf{E}$ along $\mathbf{F}$ at (1, 2, 3).
- A vector perpendicular to both $\mathbf{E}$ and $\mathbf{F}$ at (0, 1, -3) whose magnitude is unity.

3. If $\mathbf{A} = -\mathbf{a}_x + 6 \mathbf{a}_y + 5 \mathbf{a}_z$ and $\mathbf{B} = \mathbf{a}_x + 2 \mathbf{a}_y + 3 \mathbf{a}_z$. Then, find:

- The scalar projections of $\mathbf{A}$ on $\mathbf{B}$.
- The vector projection of $\mathbf{B}$ on $\mathbf{A}$.
- The unit vector perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$.